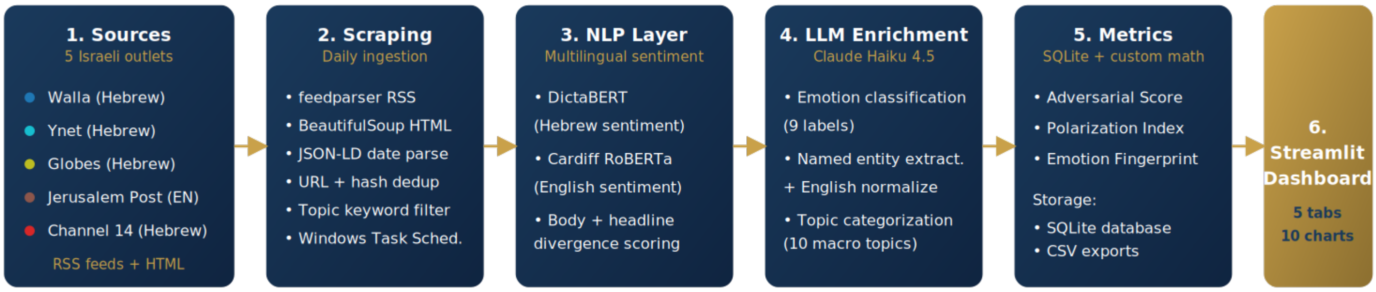
Israeli Media Framing Tracker

An end-to-end NLP pipeline that quantifies how five Israeli news outlets frame political entities and topics. Hebrew & English sentiment + LLM enrichment + three custom analytical metrics + a 5-tab Streamlit dashboard. *Portfolio analytics project. Live data collection via Windows Task Scheduler. Findings illustrate the methodology; the data window is currently ~16 days.*

Israeli Media Framing Tracker — End-to-End Pipeline



Python 3.13 · DictaBERT · Cardiff RoBERTa · Anthropic Claude · SQLite · Streamlit · Plotly

The business question

When five Israeli outlets cover the same entity or topic, how far apart are their frames, and which camp drives the gap? Four sub-questions: **(1)** Where is the line? **(2)** Who is the most polarizing figure? **(3)** Does each outlet have a distinct emotional fingerprint? **(4)** Does political lean explain the variance?

Key findings (current snapshot)

- **Polarization is structured, not noise.** Mean entity-level Polarization Index = 0.41 across 13 tracked entities (max 0.75, min 0.05).
- **Yair Lapid is the most polarizing figure.** Walla covers him at -0.667 (hostile), JPost at +0.085 (sympathetic). Netanyahu is less polarizing (0.27) because every outlet covers him adversarially.
- **Each outlet has a stable emotional fingerprint.** Ynet 42% anger; Channel 14 14% pride + 9% joy; Globes 24% anticipation + 22% fear; Walla 36% anger + 24% fear.
- **Political lean only partly explains framing.** Judicial & Legal (0.38) and Coalition & Government (0.40) follow the L/R axis; Security Operations and International Diplomacy do not. A second axis (domestic vs international focus) is at work.

Bottom line for a stakeholder

For PR & corporate comms, Globes' uniquely different topical and emotional profile makes it the entry point for financial or commercial pitches. For political consulting, the Polarization Index ranks public figures by their cross-spectrum exposure, a leading indicator of reputational risk.

Skills demonstrated

- **Multilingual NLP pipeline** (DictaBERT for Hebrew, Cardiff RoBERTa for English) on 750 articles.
- **LLM enrichment** via Anthropic Claude API for emotion, entity, and topic labels.
- **Three custom analytical metrics** designed from first principles (Adversarial Score, Polarization Index, Emotion Fingerprint).
- **5-tab Streamlit dashboard** with 10 interactive Plotly visualizations.
- **SQL data warehouse** in SQLite with deduplication and indexed lookups.
- **Production patterns**: daily ingestion via Task Scheduler, JSON-LD date extraction, backfill scripts, quality flags, structured logging.
- **Analyst judgment**: honest scoping, explicit limitations, methodology tab in-app for transparency.

Methodology, at a glance

Adversarial Score = mean(pos_score – neg_score) per group, range -1 to +1. **Polarization Index** = max – min of group means. **Emotion Fingerprint** = normalized distribution across four registers: Hostile, Anxious, Hopeful, Neutral.

Tech stack

Python 3.13 · transformers / DictaBERT / Cardiff RoBERTa · Anthropic Claude · feedparser · BeautifulSoup · SQLite via SQLAlchemy · pandas / numpy · Streamlit · Plotly · Windows Task Scheduler.

Tal Jacob · Data Analyst · **github.com/taljacob28/hebrew-news-sentiment** · taljacob28@gmail.com